

Computer Graphics (CSIT 401)

IV semester, BScCSIT

Compiled by **Ankit Bhattarai**
Ambition Guru College

Syllabus

Unit	Contents	Hours	Remarks
1.	Introduction to Computer Graphics	5	
2.	Scan Conversion Algorithms	7	<i>Very Important</i>
3.	2D Transformations and Viewing	6	<i>Very Important</i>
4.	3D Transformations and Viewing	5	
5.	3D Object Representation	6	
6.	Solid Modeling	3	
7.	Visible Surface Detection	5	
8.	Illumination, Shading and Rendering Techniques	5	
9.	OpenGL and Graphics Programming	3	

Unit 5

3D Object Representation

(6 hrs.)

Representing Surfaces: Boundary and space partitioning, Polygon Surfaces: Polygon tables, Surface normal and Spatial orientation of surfaces, Plane Equations, Polygon meshes, Wireframe Representation, Blobby Objects,

Curves: Parametric Cubic Curves, Spline Interpolation, Cubic Spline Interpolation, Hermite Curves, Bezier and B-spline curve. Quadric Surfaces: Sphere and Ellipsoid

Unit 5

Section 1: 3D Objects Representation, Polygon Surfaces

3D Object Representation

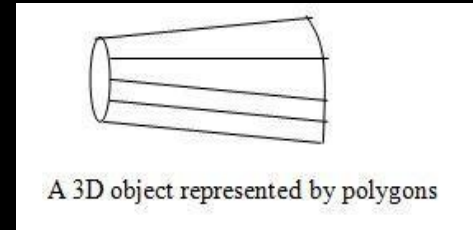
Graphics scenes can contain many different kinds of objects like trees, flowers, clouds, rocks, water etc. these cannot be described with only one methods but obviously require large precisions such as polygon & quadratic surfaces, spline surfaces, procedural methods, volume rendering, visualization techniques etc. Representation schemes for solid objects are often divided into two broad categories:

In computer graphics, **3D object representation** involves creating a digital model of a three-dimensional object that can be rendered and manipulated in a virtual environment. There are various techniques and data structures used for 3D object representation, each with its own advantages and applications.

- **Boundary representations (B-reps) or BREP:**

- describe a three-dimensional object as a set of surfaces that separate the object interior from the environment.
- B-reps describe the objects exterior. It describes a 3d object as a set of surfaces that encloses the objects interior.

Examples: *Polygon surfaces and spline patches.*



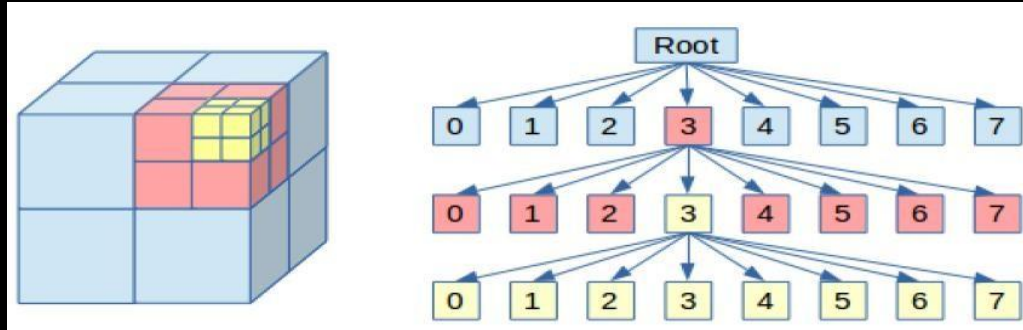
A 3D object represented by polygons

3D Object Representation

- Space-partitioning representations:

are used to describe interior properties, by partitioning the spatial region containing an object into a set of small, non- overlapping, contiguous solids (usually cubes). For example, **Octree** representation.

Space-partitioning systems are often hierarchical, meaning that a space (or a region of space) is divided into several regions, and then the same space partitioning system is recursively applied to each of the regions thus created. The regions can be organized into a tree, called a space-partitioning tree.



Polygon Surfaces

The most commonly used *boundary representation* for a three-dimensional graphics object is a set of surface polygons that enclose the object interior. Many graphics systems store all object descriptions as sets of surface polygons. This simplifies and speeds up the surface rendering and display of objects, since all surfaces are described with linear equations. For this reason, polygon descriptions are often referred to as “**standard graphics objects**”.

Generally, polygon surfaces are specified using;

1. Polygon Table
2. Polygon Meshes.
3. Plane Equations

1. Polygon Tables

Polygon Surfaces: Polygon Tables

Polygon tables can be used specified specify polygon surfaces. We specify a *polygon surface with a set of vertex coordinates and associated attribute parameters*. As information for each polygon is input, the data are placed into tables that are to be used in the subsequent processing, display, and manipulation of the objects in a scene.

Polygon tables can be organized into two groups:

1. **Geometric tables**
2. **Attribute tables**

Geometric data tables: Contain vertex coordinates and parameters to identify the spatial orientation of the polygon surfaces.

Attribute tables: Provide information for an object and includes parameters specifying the degree of transparency of the object and its surface reflectivity and texture characteristics.

Polygon Surfaces: Polygon Tables

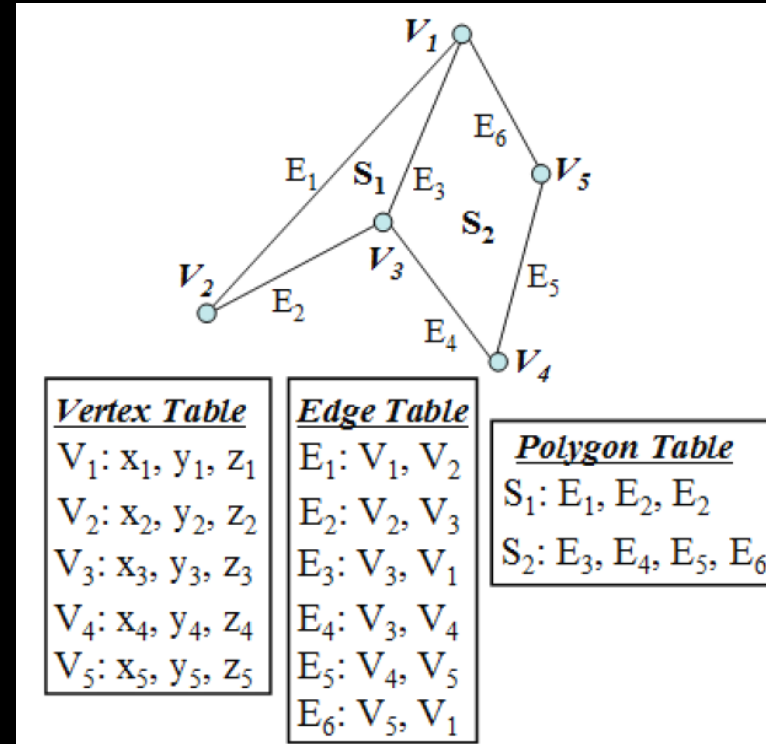
Geometric data tables contain *vertex coordinates and parameters to identify the spatial orientation of the polygon surfaces*. A convenient organization for storing geometric data is to create three lists: a vertex table, an edge table, and a polygon table.

Geometric data consists of three tables: a vertex table, an edge table, and a surface table.

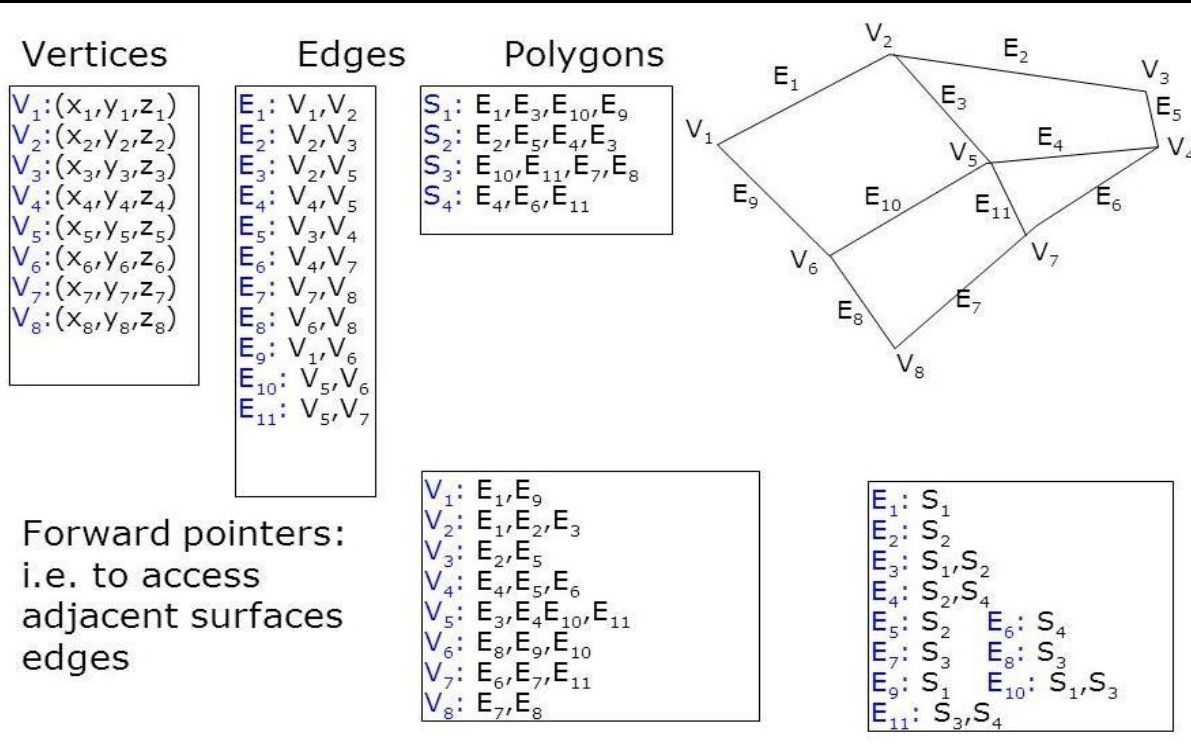
- **Vertex table:** It stores co-ordinate values for each vertex of the object.
- **Edge Table:** The edge table contains pointers back into the vertex table to identify the vertices for each polygon edge.
- **Surface table:** And the polygon table contains pointers back into the edge table to identify the edges for each polygon surfaces.

Polygon Surfaces: Polygon Tables

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Polygon Surfaces: Polygon Tables



Polygon Surfaces: Polygon Tables

Attribute tables

Attribute information for an object includes parameters specifying the degree of transparency of the object and its surface reflectivity and texture characteristics.

Degree of Transparency:

- Parameter indicating object transparency.
- Represented by an alpha value (0 to 1).
- 0 is fully transparent, 1 is fully opaque.

Surface Reflectivity:

- Describes how much light reflects off the object's surface.
- Specified by values between 0 (non-reflective) and 1 (fully reflective).

Texture Characteristics:

- Includes color, pattern, and bumpiness on the object's surface.
- Texture mapping involves applying 2D images to simulate surface details.

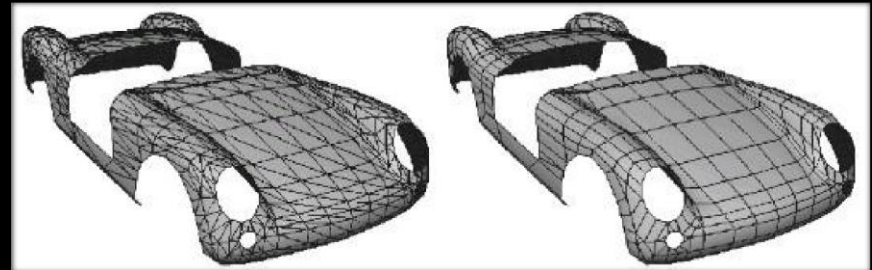
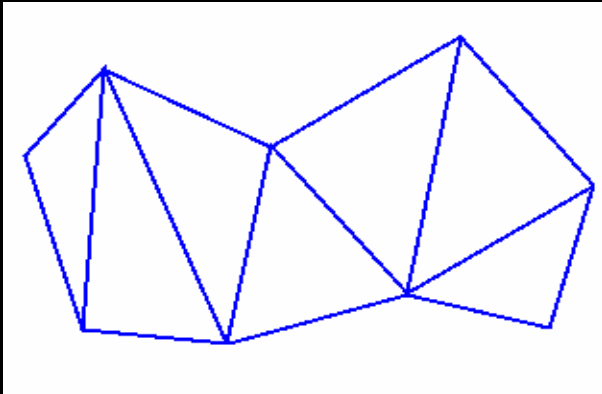
2. Polygon Meshes

2. Polygon Meshes

A polygon mesh is a collection of vertices, edges and faces that defines the shape of a polyhedral object in 3D computer graphics

1. Triangular Mesh :

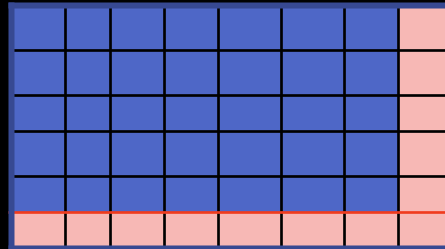
It produces $n - 2$ connected triangles, given the coordinates for n vertices.



2. Polygon Meshes

2 . Quadrilateral Mesh

Another similar functions the quadrilateral mesh that generates a mesh of $(n-1)(m-1)$ quadrilaterals given that coordinates for an n by m array of vertices.



6 by 8 vertices array , 35
element quadrilateral
mesh

If the surface of 3D object is planner. It is comfortable to represent surface with meshes

2. Polygon Meshes

- Representation Polygon meshes

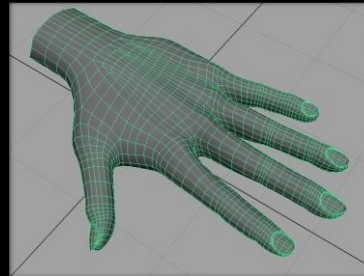
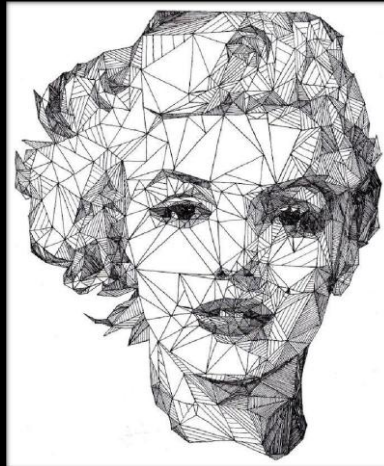
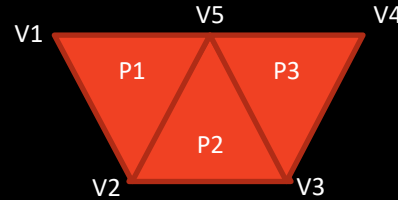
$$P = \{ (X_1, Y_1, Z_1), (X_2, Y_2, Z_2), \dots, (X_n, Y_n, Z_n) \}$$

Each polygon represent by a list of vertex of coordinate

$$P1 = \{ V1, V2, V3 \}$$

$$P2 = \{ V2, V3, V5 \}$$

$$P3 = \{ V3, V4, V5 \}$$



3. Plane Equation

3. Plane Equation

- Plane equation method is the method for representing polygon surface for 3D object. The information about spatial orientation of object is described by its individual surface which is obtained by vertex coordinate values & equation of each plane gives a description of each surface.
- It is used to determine the spatial orientation of the individual surface component of the object. The equation for a plane surface can be expressed in the form

$$Ax + By + Cz + D = 0$$

where (x, y, z) is any point on the plane, and the coefficients A , B , C , and D are constants. Let (x_1, y_1, z_1) , (x_2, y_2, z_2) , and (x_3, y_3, z_3) be three successive polygon vertices of the polygon.

$$Ax_1 + By_1 + Cz_1 + D = 0,$$

$$Ax_2 + By_2 + Cz_2 + D = 0,$$

$$Ax_3 + By_3 + Cz_3 + D = 0$$

3. Plane Equation

- Using Cramer's Rule

$$Ax_1 + By_1 + Cz_1 + D = 0,$$

$$Ax_2 + By_2 + Cz_2 + D = 0,$$

$$Ax_3 + By_3 + Cz_3 + D = 0$$

$$A = \begin{vmatrix} 1 & y_1 & z_1 \\ 1 & y_2 & z_2 \\ 1 & y_3 & z_3 \end{vmatrix} \quad B = \begin{vmatrix} x_1 & 1 & z_1 \\ x_2 & 1 & z_2 \\ x_3 & 1 & z_3 \end{vmatrix} \quad C = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \quad D = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$$

Expanding the determinant, we can write that,

$$A = y_1(z_2 - z_3) + y_2(z_3 - z_1) + y_3(z_1 - z_2)$$

$$B = z_1(x_2 - x_3) + z_2(x_3 - x_1) + z_3(x_1 - x_2)$$

$$C = x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)$$

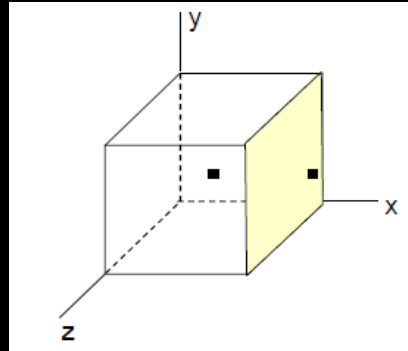
$$D = -x_1(y_2 z_3 - y_3 z_2) - x_2(y_3 z_1 - y_1 z_3) - x_3(y_1 z_2 - y_2 z_1)$$

3. Plane Equation

- Inside outside tests of the surface:

$Ax + By + Cz + D < 0$, point (X,Y,Z) is inside the surface

$Ax + By + Cz + D > 0$, point (X,Y,Z) is outside the surface



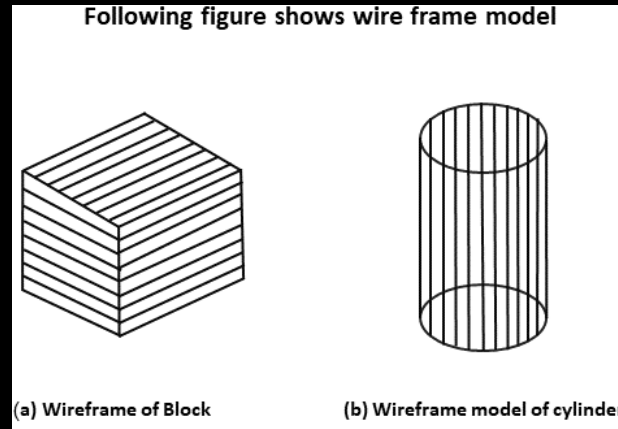
Unit 5

Section 2: Wireframe, Surface Normal and Spatial Orientation of Surfaces, Blobby Objects

Wireframe

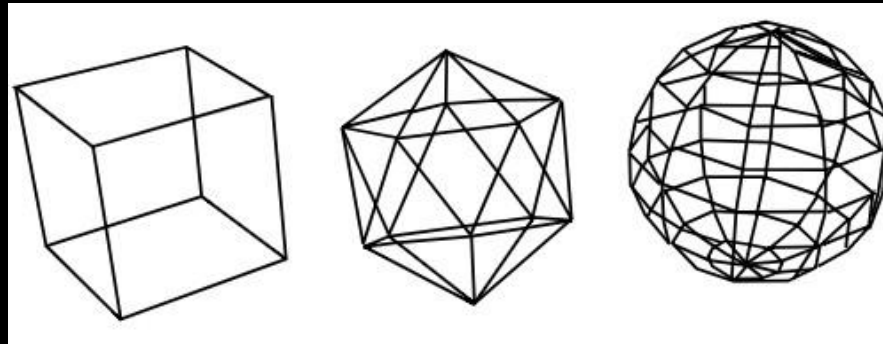
Wireframe

- A wireframe is a *three-dimensional model that only includes vertices and lines*. It does not contain surfaces, textures, or lighting like a 3D mesh.
- Instead, a wireframe model is a 3D image comprised of only "wires" that represent three-dimensional shapes.
- A **wire-frame model** is a visual presentation of a 3 - dimensional (3D) or physical object used in 3D computer graphics.



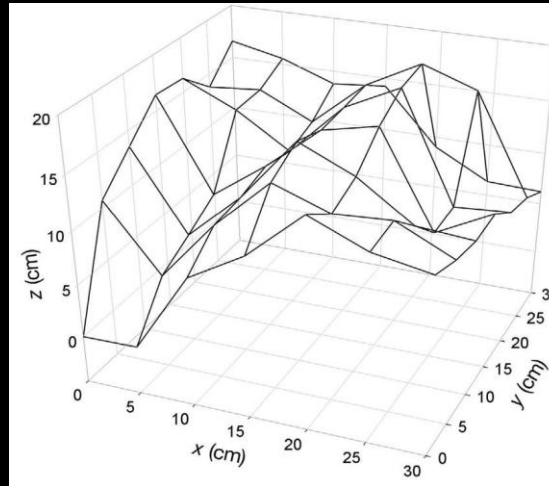
Wireframe

- Wireframes provide the most basic representation of a three- dimensional scene or object.
- They are often used as the starting point in 3D modeling since they create a "frame" for 3D structures. For example, a 3D graphic designer can create a model from scratch by simply defining points (vertices) and connecting them with lines (paths).
- Once the shape is created, surfaces or textures can be added to make the model appear more realistic.



Wireframe

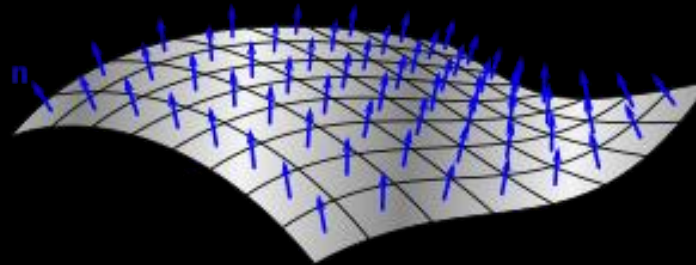
- The lines within a wireframe connect to create polygons, such as triangles and rectangles, that together represent three-dimensional shapes.
- The result may be as simple as a cube or as complex as a three-dimensional scene with people and objects. The number of polygons within a wireframe is typically a good indicator of how detailed the 3D model is.



Surface Normal and Spatial Orientation of Surfaces

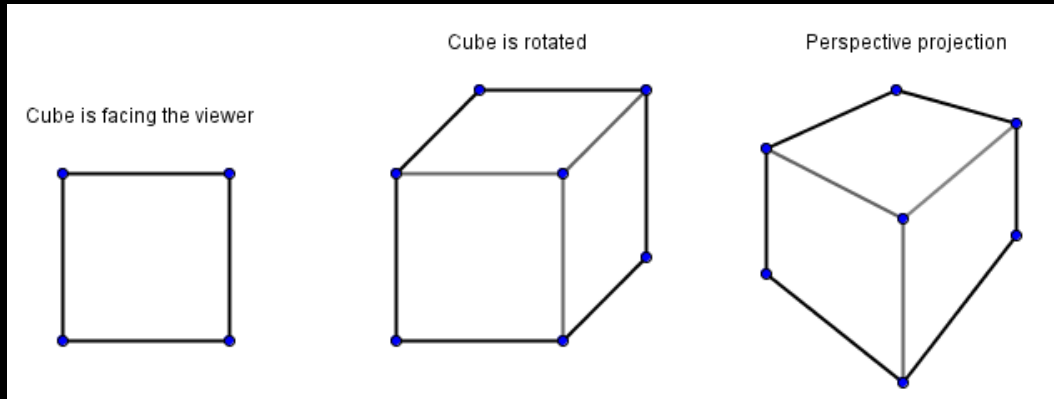
Surface Normal

- The surface normal *at a point on an object is a vector that is perpendicular (or orthogonal) to the surface at that point.*
- It indicates the direction in which the surface is facing.
- The surface normal is crucial for lighting calculations in computer graphics, as it helps determine how light interacts with the surface.



Spatial Orientation of Surfaces

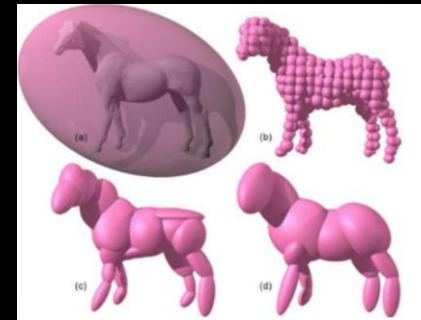
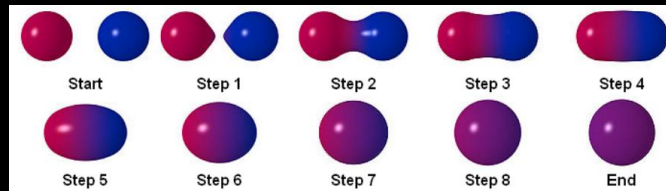
- Spatial orientation refers to the *positioning of an object or surface in three-dimensional space*.
- For surfaces, this involves understanding how the surface is oriented relative to the coordinate axes (X, Y, Z).
- The spatial orientation is often described by the angles between the surface normal and the coordinate axes.



Blobby Objects

Bloppy Objects

- Bloppy objects are the objects which are non-rigid and do not retain their fixed size. Bloppy objects (as is obvious by their name) are the objects which change their shape and size on the basis of their states.
- By a bloppy object we mean a nonrigid object. That is things, like cloth, rubber, liquids, water droplets, etc.
- These objects tend to exhibit a degree of fluidity.
- For example, in a chemical compound electron density clouds tend to be distorted by the presence of other atoms/molecules.



More Blobby Objects

- Several models have been developed to handle these kind of objects.
- One technique is to use a combination of Gaussian density functions (Gaussian bumps).



- A surface function could then be defined by:

$$f(x,y,z) = \sum_k b_k \cdot \exp(-a_k \cdot r_k^2) - T = 0$$

where $r_k^2 = x_k^2 + y_k^2 + z_k^2$

- Another technique called the **meta-ball** technique is to describe the object as being made of density functions much like balls.
- The advantage here is that the density function falls off in a finite interval.

Unit 5

Section 3: Curves, Spline Representation

Curves

- In the field of computer graphics, it's common to depict various objects on a screen. These objects aren't always two-dimensional; frequently, we must draw curves multiple times to accurately represent their shapes.
- In computer graphics, a **curve refers to a continuous and smooth line that represents a mathematical function or a path between two or more points.**
- Curves are commonly used to depict various shapes, such as arcs, circles, and complex contours, by defining a set of points through mathematical equations or algorithms. They play a crucial role in creating realistic and visually appealing graphics by enabling the representation of smooth and intricate shapes on digital displays.

Curves: Explicit Curves

Curves can be broadly classified into three categories - **explicit**, **implicit**, and **parametric curves**.

1. Explicit curves: These are defined by an equation where one variable (usually y) is expressed explicitly as a function of the other variable (usually x). In simpler terms, the equation directly gives you the y -coordinate for any given x -coordinate.

- ✓ For example, the equation of a line with slope m and y -intercept b is $y = mx + b$.
Here, y is explicitly defined in terms of x .

Curves: Implicit Curves

2. Implicit curves: These are defined by an equation where both variables (x and y) are present, but the equation is not solved for either variable.

In other words, the relationship between x and y is described indirectly, and you cannot readily isolate y to get an explicit function.

- ✓ For example, the *equation of a circle with center (h, k) and radius r is $(x - h)^2 + (y - k)^2 = r^2$* . Here, x and y are intertwined in the equation, and it's not directly possible to express y solely in terms of x .

Curves: Parametric Curves

3. Parametric curves:

- ✓ Curves having *parametric form* are called parametric curves. The explicit and implicit curve representations can be used only when the function is known. In practice the parametric curves are used.
- ✓ These are defined by a set of equations where both x and y are expressed as functions of a single parameter (often denoted by t).

A two-dimensional parametric curve has the following form:

$$P(t) = f(t), g(t) \text{ or } P(t) = x(t), y(t)$$

The functions f and g become the x, y coordinates of any point on the curve, and the points are obtained when the parameter t is varied over a certain interval $[a, b]$, normally $[0, 1]$.

Curves: Parametric Curves

Another case: the parametric equations for a circle with center (h, k) and radius r are:

$$x = h + r * \cos(t)$$

$$y = k + r * \sin(t)$$

Here, x and y are defined in terms of the parameter t , which can vary, for instance, from 0 to 2π to trace out the entire circle.

Why curves are important in Computer Graphics ?

- ✓ **Smooth Surfaces:** Curves enable the creation of smooth and continuous surfaces, contributing to the realistic representation of objects in computer-generated images.
- ✓ **Realistic Shapes:** Many real-world objects have curved or intricate shapes that can be accurately depicted using curves. This enhances the realism of computer-generated scenes.
- ✓ **Animation and Motion:** Parametric curves are commonly used to represent paths of motion in animations. By defining curves, animators can smoothly transition objects or characters through various positions.
- ✓ **Bezier and Splines for Precision:** Bezier curves and splines provide precise control over the shape of curves, allowing designers and animators to create complex and aesthetically pleasing graphics with ease.
- ✓ **Interpolation:** Curves are essential for interpolating between keyframes in animations, ensuring a smooth transition of positions, orientations, and other attributes over time.

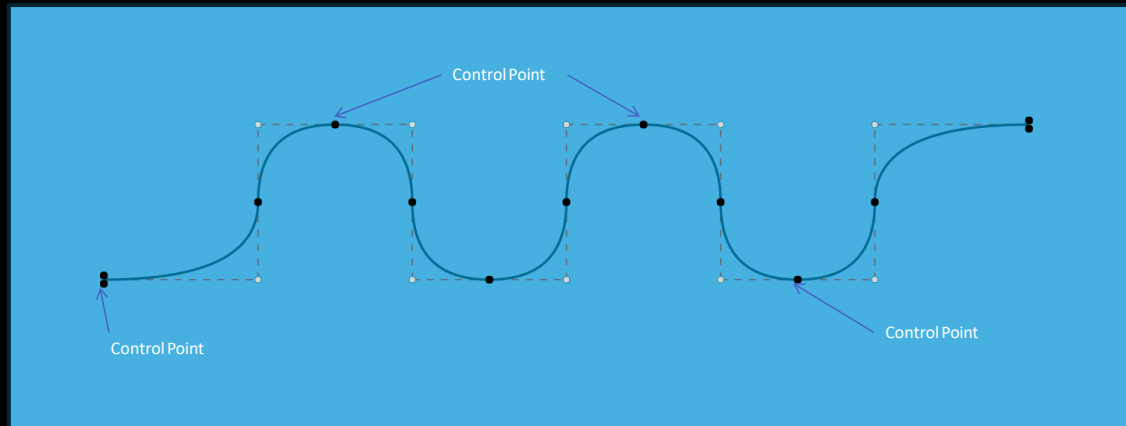
Why curves are important in Computer Graphics ?

- ✓ **Fonts and Typography:** Curves are extensively used to define the shapes of characters in fonts. They contribute to the creation of aesthetically pleasing and readable text in computer graphics applications.
- ✓ **Vector Graphics:** Curves are fundamental in vector graphics, where images are defined as a series of mathematical curves rather than pixels. This allows for scalability without loss of quality.
- ✓ **CAD and Design:** In Computer-Aided Design (CAD), curves are used to represent precise shapes and contours, aiding engineers and designers in the creation of detailed models.
- ✓ **Mathematical Modeling:** Curves provide a mathematical foundation for modeling complex shapes and surfaces, allowing for the representation of a wide range of objects in virtual environments.
- ✓ **User Interface Design:** Curves are used in designing graphical user interfaces to create smooth transitions, aesthetically pleasing layouts, and visually appealing elements.

Spline Representation

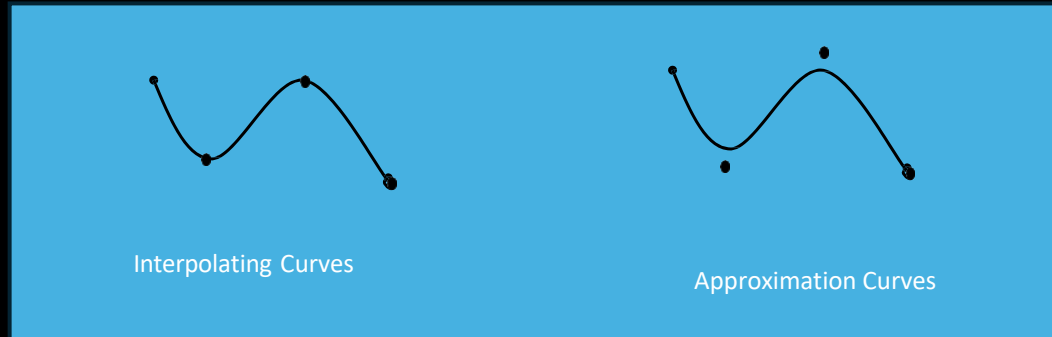
Spline Representation

- A spline curve is a mathematical representation for which it is easy to build an interface that will allow a user to design and control the shape of complex curves and shapes
- The general approach is that the user enters a sequence of points and a curve is constructed whose shape closely follows this sequence. The points are called **control point**.

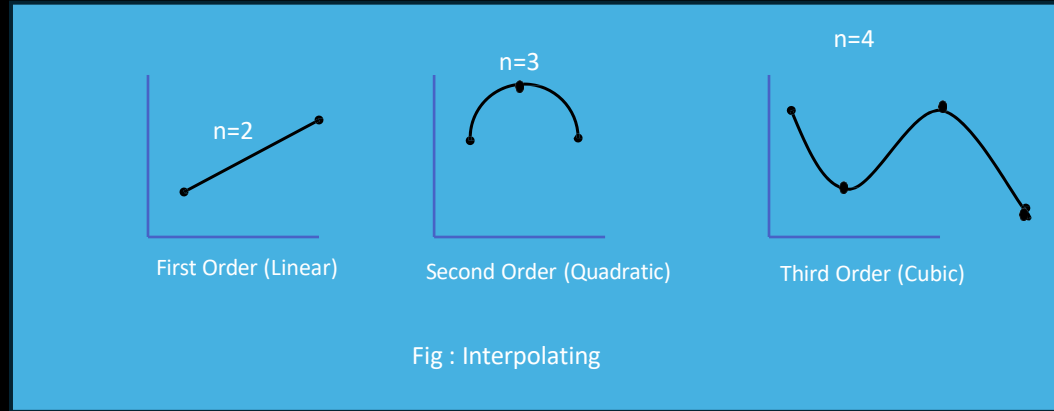


Spline Representation

- A curve is actually passes through each control point is called **interpolating curve**
- A curve that passes near to the control point but not necessarily through them is called an **approximating curve**.



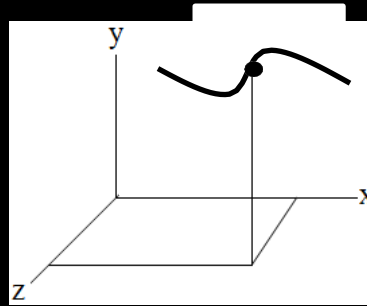
Spline Representation



Parametric Cubic Curve

- A parametric cubic curve is defined as

$$P(t) = \sum_{i=0}^3 a_i t^i \quad 0 \leq t \leq 1 \quad \text{----- (i)}$$



Where, $P(t)$ is a point on the curve
 a = algebraic coefficients
 t = tangent Vector

Parametric Cubic Curve

Expanding equation (i) yield

$$P(t) = a_3 t^3 + a_2 t^2 + a_1 t + a_0 \text{----- (ii)}$$

This equation is separated into three components of $P(t)$

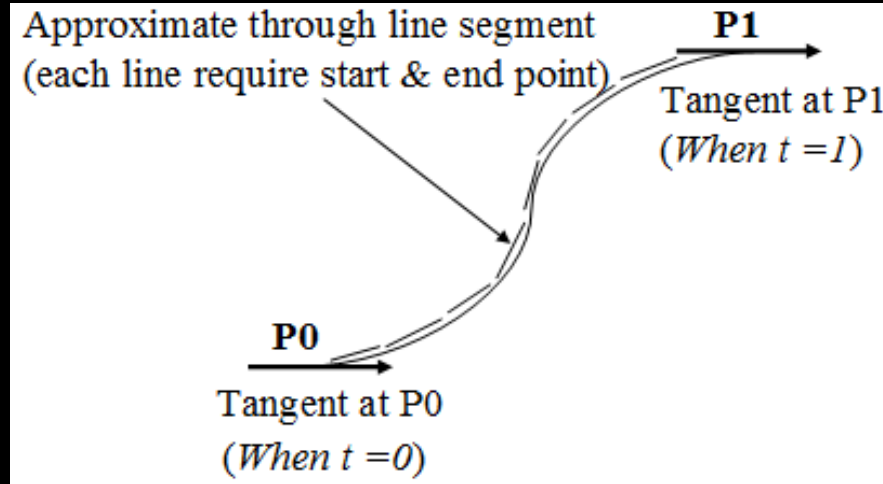
$$x(t) = a_{3x} t^3 + a_{2x} t^2 + a_{1x} t + a_{0x}$$

$$y(t) = a_{3y} t^3 + a_{2y} t^2 + a_{1y} t + a_{0y}$$

$$z(t) = a_{3z} t^3 + a_{2z} t^2 + a_{1z} t + a_{0z} \text{----- (iii)}$$

- To be able to solve (iii) the **twelve unknown** coefficients a_{ij} (Algebraic coefficients) must be specified
- From the known end point coordinates of each segment, six of the twelve needed equations are obtained.
- The other six are found by using tangent vectors at the two ends of each segment
- The direction of the tangent vectors establishes the slopes(direction cosines) of the curve at the end point

Parametric Cubic Curve



Parametric Cubic Curve

- This procedure for defining a cubic curve using **end points and tangent vector** is one form of **Hermite interpolation**
- Each cubic curve segment is parameterized from 0 to 1 so that known end points correspond to the limit values of the parametric variable t , that is $P(0)$ and $P(1)$
- Substituting $t = 0$ and $t = 1$ the relationship between two end point vectors and the algebraic coefficients are found

$$P(t) = a_3 t^3 + a_2 t^2 + a_1 t + a_0$$

$$P(0) = a_0$$

$$P(1) = a_3 + a_2 + a_1 + a_0 \text{ ---- (IV)}$$

Parametric Cubic Curve

- To find the tangent vectors equation (ii) must be differentiated with respect to t

$$P(t) = a_3 t^3 + a_2 t^2 + a_1 t + a_0$$

$$P'(t) = 3 a_3 t^2 + 2 a_2 t + a_1$$

- The tangent vectors at the two end points are found by substituting $t = 0$ and $t = 1$ in this equation

$$P'(0) = a_1 \qquad P'(1) = 3 a_3 + 2 a_2 + a_1 \text{-----} (V)$$

Parametric Cubic Curve

- The algebraic coefficients 'a_i' in equation (ii) can now be written explicitly in terms of boundary conditions – endpoints and tangent vectors are

- $a_0 = P(0)$ $a_1 = P'(0)$
- $a_2 = -3P(0) - 3P(1) - 2P'(0) - P'(1)$
- $a_3 = 2P(0) - 2P(1) + P'(0) + P'(1)$

(Note: - The value of a₂ & a₃ can be determined by solving the equation IV & V)

- Substituting these values of 'a_i' in equation (ii) and rearranging the terms yields

$$P(t) = (2t^3 - 3t^2 + 1) P(0) + (-2t^3 + 3t^2) P(1) + (t^3 - 2t^2 + t) P'(0) + (t^3 - t^2) P'(1)$$

- The values of P(0), P(1), P'(0), P'(1) are called *geometric coefficients* and represent the known vector quantities in the above equation
- The polynomial coefficients of these vector quantities are commonly known as *blending functions*. By varying parameter t in these blending function from 0 to 1 several points on curve segments can be found.

Unit 5

Section 4: Bezier Curve, Numerical

Bezier Curve

Bezier Curve and Surfaces

Bezier splines are highly useful, easy to implement and convenient for curve and surface design so are widely available in various CAD systems, graphics packages, drawing and painting packages.

Bezier curve

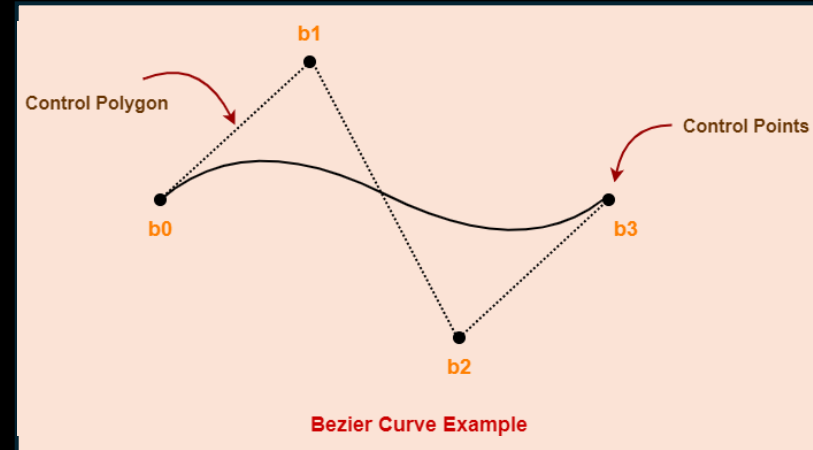
In general, a Bezier curve can be fitted to any number of control points. The number of control points to be *approximated* and their relative position determine the degree of the Bezier polynomial. As with the interpolation splines, a Bezier curve can be specified with boundary conditions, with a characterizing matrix, or with *blending functions*.

✓ **Bezier Curve** is parametric curve defined by a set of control points.

- Two points are ends of the curve.
- Other points determine the shape of the curve.

Here,

- This Bezier curve is defined by a set of control points b_0 , b_1 , b_2 and b_3 .
- Points b_0 and b_3 are ends of the curve.
- Points b_1 and b_2 determine the shape of the curve.



Bezier Curve and Surfaces

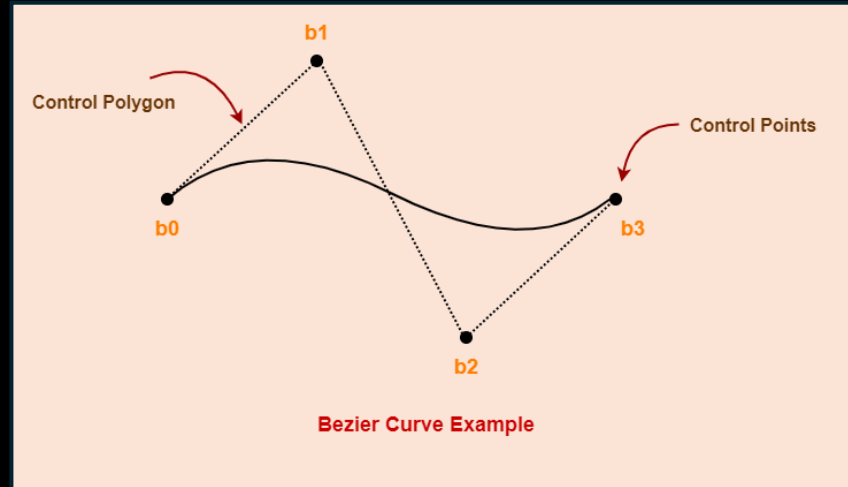
The Bezier curve has two important properties:

1. It always passes through the first and last control points.
2. It lies within the convex hull (convex polygonal boundary) of the control points. This follows from the properties of Bezier blending function: they are positive and their sum is always 1, i.e.

$$\sum_{k=0}^n z_k \text{BEZ}_{k,n}(u) = 1$$

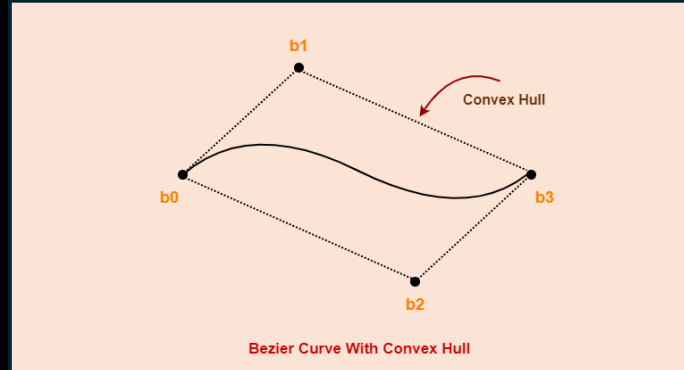
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- Points b_0 and b_3 are ends of the curve.
- Points b_1 and b_2 determine the shape of the curve.

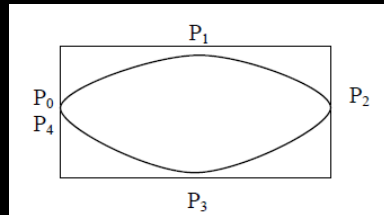


Properties Of Bezier Curve

1. Bezier curve lies in the convex hull of the control points which ensure that the curve smoothly follows the control Points



2. Four Bezier polynomials are used in the construction of curve to fit four control points
3. It always passes through the end points.
4. Closed curves can be generated by specifying the first and last control points at the same position.



Properties Of Bezier Curve

5. Depending on the positioning of control points, Bezier curves can produce a wide variety of shapes, from simple linear transitions to complex and curved paths.
6. The tangent to the curve at an end point is along the line joining the end point to the adjacent control point.
7. Bezier curves are typically represented parametrically. The position of a point on the curve is determined by a parameter (often denoted as t), ranging from 0 to 1.
8. The degree of a Bezier curve is determined by the number of control points minus one. For example, a cubic Bezier curve has four control points.
9. The order of the polynomial defining the curve segment is equal to the total number of control points.

Application Of Bezier Curve

Bezier curves have their applications in the following fields-

1. Computer Graphics:

- Bezier curves are widely used in computer graphics to model smooth curves.
- The curve is completely contained in the convex hull of its control points.
- So, the points can be graphically displayed & used to manipulate the curve intuitively.

2. Animation:

- Bezier curves are used to outline movement in animation applications such as Adobe Flash and synfig.
- Users outline the wanted path in Bezier curves.
- The application creates the needed frames for the object to move along the path.
- For 3D animation, Bezier curves are often used to define 3D paths as well as 2D curves.

3. Fonts:

- True type fonts use composite Bezier curves composed of quadratic Bezier curves.
- Modern imaging systems like postscript, asymptote etc. use composite Bezier curves composed of cubic Bezier curves for drawing curved shapes.

Bezier Curve

Suppose we are given $n + 1$ control-point positions:

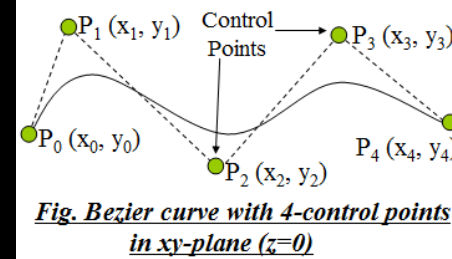
$\mathbf{p}_k = (x_k, y_k, z_k)$, with k varying from 0 to n . These coordinate points can be blended to produce the following position vector $\mathbf{P}(u)$, which describes the path of an approximating Bezier polynomial function between $\mathbf{p}_0$ and $\mathbf{p}_n$.

$$\mathbf{P}(u) = \sum_{k=0}^n \mathbf{p}_k \text{BEZ}_{k,n}(u), \quad 0 \leq u \leq 1$$

Bezier Curve

This, vector equation represents a set of three parametric equations for the individual curve coordinates:

$$\begin{aligned} x(u) &= \sum_{k=0}^n x_k \text{BEZ}_{k,n}(u) \\ y(u) &= \sum_{k=0}^n y_k \text{BEZ}_{k,n}(u) \\ z(u) &= \sum_{k=0}^n z_k \text{BEZ}_{k,n}(u) \end{aligned}$$

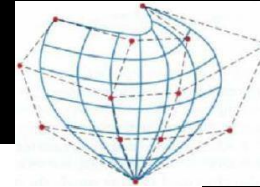


The Bezier blending functions $\text{BEZ}_{k,n}(u)$ are the Bernstein polynomials:

$$\text{BEZ}_{k,n}(u) = C(n, k) u^k (1-u)^{n-k}$$

Where the $C(n, k)$ are the binomial coefficients:

$$C(n, k) = \frac{n!}{k! (n-k)!}$$



Bezier Curve

Now

$$P(u) = \sum_{k=0}^n p_k \text{BEZ}_{k,n}(u), \quad 0 \leq u \leq 1$$

($n = 3$)

Then,

$$P(u) = P_0 \text{BEZ}_{0,3}(u) + P_1 \text{BEZ}_{1,3}(u) + P_2 \text{BEZ}_{2,3}(u) + P_3 \text{BEZ}_{3,3}(u)$$

Four blending functions must be found based on Bernstein Polynomials

$$\begin{aligned} \text{BEZ}_{0,3}(u) &= \frac{3!}{0! 3!} u^0 (1-u)^3 = (1-u)^3 & \text{BEZ}_{1,3}(u) &= \frac{3!}{1! 2!} u^1 (1-u)^2 = 3u(1-u)^2 \\ \text{BEZ}_{2,3}(u) &= \frac{3!}{2! 1!} u^2 (1-u) = 3u^2(1-u) & \text{BEZ}_{3,3}(u) &= \frac{3!}{3! 0!} u^3 (1-u)^0 = u^3 \end{aligned}$$

Bezier Curve

Normalizing properties apply to blending functions that means they all add up to one.

Substituting these functions in above equation

$$P(u) = (1-u)^3 P_0 + 3u(1-u)^2 P_1 + 3u^2(1-u) P_2 + u^3 P_3$$

When $u = 0$ then $P(u) = P_0$ and

when $u = 1$ then $P(u) = P_3$

in Matrix Form

$$Q(u) = \begin{bmatrix} (1-u)^3 & 3u(1-u)^2 & 3u^2(1-u) & u^3 \end{bmatrix} \begin{bmatrix} P_0 \\ P_1 \\ P_2 \\ P_3 \end{bmatrix}$$

Bezier Curve Numerical

Construct the Bezier curve with the following polygon vertices (control points):
 $A(1,1)$, $B(2,3)$, $C(4,3)$, and $D(6,2)$.

Or,

Consider the Bezier Curve of order 3 and with 4 polygon vertices $A(1,1)$, $B(2,3)$, $C(4,3)$, and $D(6,2)$. Determine any 5 points lying on the curve. Also draw the rough sketch of the curve.

Bezier Curve Numerical



AMBITION GURU

Consider the Bezier Curve of order 3 and with 4 polygon vertices A(1,1), B(2,3), C(4,3), and D(6,2). Determine any 5 points lying on the curve. Also draw the rough sketch of the curve.

Solution:

$$P(u) = (1-u)^3 P_0 + 3u(1-u)^2 P_1 + 3u^2(1-u) P_2 + u^3 P_3$$

$$P(u) = [1 \ 1] (1-u)^3 + 3u(1-u)^2 [2 \ 3] + 3u^2(1-u) [4 \ 3] + u^3 [6 \ 2]$$

①

To get 5 points lying on the curve, assume any 5 values of u lying in the range $0 \leq u \leq 1$.

Let 5 values of u are: $0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$.

0, 0.2, 0.5, 0.7, 1

Can be used

Bezier Curve Numerical



AMBITION GURU

Consider the Bezier Curve of order 3 and with 4 polygon vertices A(1,1), B(2,3), C(4,3), and D(6,2). Determine any 5 points lying on the curve. Also draw the rough sketch of the curve.

For $u=0$ in eqn ①, we get:

$$P(0) = [1 \ 1] (1-0)^3 + 0 + 0 + 0 \\ = [1 \ 1]$$

For $u = \frac{1}{4}$ in eqn ①, we get:

$$P\left(\frac{1}{4}\right) = [1 \ 1] \left(1 - \frac{1}{4}\right)^3 + 3 \cdot \frac{1}{4} \left(1 - \frac{1}{4}\right)^2 [2 \ 3] + 3 \cdot \left(\frac{1}{4}\right)^2 \left(1 - \frac{1}{4}\right) [4 \ 3] \\ + \left(\frac{1}{4}\right)^3 \cdot [6 \ 2]$$

Bezier Curve Numerical



AMBITION GURU

$$P(u) = (1-u)^3 P_0 + 3u(1-u)^2 P_1 + 3u^2(1-u) P_2 + u^3 P_3$$

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$$= \frac{27}{64} [1, 1] + \frac{27}{64} [2, 3] + \frac{9}{64} [4, 3] + \frac{1}{64} [6, 2]$$

Q

$$x(u) = \frac{27}{64} \times 1 + \frac{27}{64} \times 2 + \frac{9}{64} \times 4$$

NOTE:

(1, 1)

(2, 3)

$$+ \frac{1}{64} \times 6 = \frac{123}{64} = 1.9218$$

$x_0 y_0$

$x_1 y_1$

$$y(u) = \frac{27}{64} \times 1 + \frac{27}{64} \times 3 + \frac{9}{64} \times 3$$

(4, 3)

(6, 2)

$x_2 y_2$

$x_3 y_3$

$$+ \frac{1}{64} \times 2 = \frac{137}{64} = 2.14$$

$$= (1.9218, 2.14)$$

Bezier Curve Numerical



AMBITION GURU

For $u = \frac{1}{2}$ in eqn ①, we get

$$P\left(\frac{1}{2}\right) = \left(1 - \frac{1}{2}\right)^3 [1 \ 1] + 3 \cdot \frac{1}{2} \cdot \left(1 - \frac{1}{2}\right)^2 [2 \ 3] + \frac{3}{8}$$

$$3 \cdot \left(\frac{1}{2}\right)^2 \cdot \left(1 - \frac{1}{2}\right) [4 \ 3] + \left(\frac{1}{2}\right)^3 [6 \ 2]$$

$$= \frac{1}{8} \cdot [1 \ 1] + \frac{3}{8} \cdot [2 \ 3] + \frac{3}{8} [4 \ 3] + \frac{1}{8} \cdot [6 \ 2]$$

$$= \left(\frac{1}{8} \times 1 + \frac{3}{8} \times 2 + \frac{3}{8} \times 4 + \frac{1}{8} \times 6, \frac{1}{8} \times 1 + \frac{3}{8} \times 3 + \frac{3}{8} \times 3 + \frac{1}{8} \times 2 \right)$$

$$= \left(\frac{25}{8}, \frac{21}{8} \right) = (3.125, 2.625)$$

Bezier Curve Numerical



AMBITION GURU

For $u = \frac{3}{4}$ in eqn ①, we get

$$P\left(\frac{3}{4}\right) = \left(1 - \frac{3}{4}\right)^3 P_0 + 3 \cdot \frac{3}{4} \cdot \left(1 - \frac{3}{4}\right)^2 \cdot P_1 + 3 \cdot \left(\frac{3}{4}\right)^2 \cdot \left(1 - \frac{3}{4}\right) P_2 + \left(\frac{3}{4}\right)^3 P_3$$

$$= \frac{1}{64} [1 \ 1] + \frac{9}{64} [2 \ 3] + \frac{27}{64} [4 \ 3] + \frac{27}{64} [6 \ 2]$$

$$= \left(\frac{1}{64} \times 1 + \frac{9}{64} \times 2 + \frac{27}{64} \times 4 + \frac{27}{64} \times 6, \frac{1}{64} \times 1 + \frac{9}{64} \times 3 + \frac{27}{64} \times 3 + \frac{27}{64} \times 2 \right)$$

$$= \left(\frac{289}{64}, \frac{163}{64} \right) = (4.5156, 2.5468)$$

For $u = 1$ in eqn ①, we get

$$P(1) = 0 + 0 + 0 + (1)^3 \cdot [6 \ 2]$$

$$\therefore P(1) = [6 \ 2]$$

Now, draw the graph of the above points.

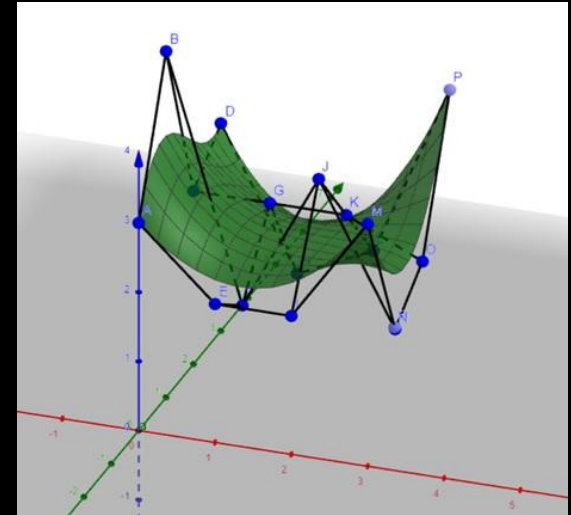
Unit 5

Section 4: Bezier Surfaces, Quadratic Surfaces

Bezier Surfaces

Bezier Surfaces

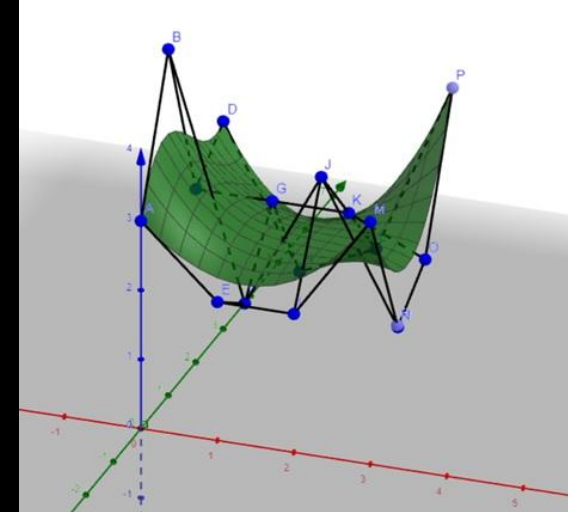
- A Bezier surface is a mathematical representation used in **computer graphics to create smooth surfaces**. It is defined by a grid of control points, and the surface is shaped according to these points, similar to how Bezier curves are used for smooth lines.
- Bezier surfaces are a type of mathematical spline used in computer graphics, computer-aided design, and finite element modeling.
- Specifically, a Bezier surface is formed as the Cartesian product of the blending functions of two orthogonal Bezier curves, allowing for complex surface modeling.



Bezier Surfaces

Key Characteristics

- **Control Points:** Defined by a grid of control points, often called a Bezier net.
- **Tensor Product:** Most surfaces are "tensor product" surfaces, essentially created by "multiplying" two orthogonal Bezier curves.
- **Patch-Based:** The most common form is the bicubic Bezier patch, which uses a grid of 16 control points to provide high flexibility for design.



Quadric Surfaces

Quadric surfaces

- If a surface is the graph in three-space of an equation of **second degree**, it is called a quadric surface. Cross section of quadric surface are conics.
- A quadric surface is a type of surface in three-dimensional space that can be defined algebraically by a second-degree equation.
- Two common examples of quadric surfaces are spheres and ellipsoids.
- Quadric Surface is one of the frequently used 3D objects surface representation. The quadric surface can be represented by a **second degree** polynomial. This includes:

Quadric surfaces

1. Sphere: For the set of surface points (x,y,z) the spherical surface is represented as:

$$x^2+y^2+z^2=r^2, \text{ with radius } r \text{ and centered at co-ordinate origin.}$$

2. Ellipsoid: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, where (x,y,z) is the surface points and a,b,c are the radii on X,Y and Z directions respectively.

3. Elliptic paraboloid: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = z$

4. Hyperbolic paraboloid: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = z$

5. Elliptic cone: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$

6. Hyperboloid of one sheet: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$

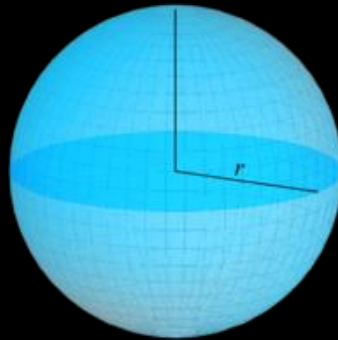
7. Hyperboloid of two sheet: $\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$

Quadric surfaces: Sphere

- A sphere is a set of all points in three-dimensional space that are a fixed distance, called the radius, from a given point called the center.
- Mathematically, a sphere with center (h,k,l) and radius r can be represented by the equation:

$$(x-h)^2+(y-k)^2+(z-l)^2 = r^2$$

- This equation describes all points (x,y,z) that are equidistant from the center (h,k,l) by a distance of r .



Quadric surfaces: Ellipsoid

- An ellipsoid is a generalization of a sphere, where the radii along each axis may be different. It's a surface where each cross-section parallel to one of the coordinate planes is an ellipse.
- The equation of an ellipsoid centered at the origin (0,0,0) with semi-axes lengths a , b , and c along the x , y , and z axes respectively is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Questions

1. What is a polygon mesh? Describe its types.
2. Explain Bezier curve and its properties.
3. Construct the Bezier curve with the following polygon vertices (control points): A(1,1), B(2,3), C(4,3), and D(6,2).
4. Explain polygon surface representation using Polygon table and polygon meshes.
5. What are blobby objects? How is it represented? Explain the wireframe representation of 3D objects.
6. Explain about parametric curve. Describe the properties of Bezier curve.
7. Explain the Bezier curve and B-Spline curve.
8. Define blobby objects. How is polygon table used in representing polygons?
9. Explain quadratic surfaces in brief.

THANK YOU
Any Queries ?